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Shri Vile Parle Kelavani Mandal'S
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Empowered Autonomous) Affiliated to the University of Mumbai

Department of Information Technology

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COURSE NAME: Computer Networks **CLASS: T. Y. B.Tech (I3-1)**

ROLL_NO: I064

Experiment No. 7

CO/LO: Describe the fundamental components of computer networks and layered reference models.

Aim: Use Wireshark to understand the operation of TCP/IP layers :

- **Ethernet Layer : Frame header, Frame size etc.**
- **Data Link Layer : MAC address, ARP (IP and MAC address binding)**
- **Network Layer : IP Packet (header, fragmentation), ICMP (Query and Echo)**
- **Transport Layer: TCP Ports, TCP handshake segments etc.**
- **Application Layer: DHCP, FTP, HTTP header formats**

EXERCISE

Description

This experiment demonstrates the use of **Wireshark**, a network protocol analyzer, to capture and analyze packets transmitted over a network. The captured packets are examined layer-by-layer to understand how data is encapsulated and exchanged between a client and a server.

Wireshark displays information such as source and destination addresses, protocols, packet length, TCP ports, HTTP requests and responses, DNS queries, ICMP messages, and packet headers. The **Follow TCP Stream** feature is also used to reconstruct and inspect the data exchanged between two endpoints.

The experiment focuses on understanding the operation of the TCP/IP protocol stack, including Ethernet frames, MAC and IP addressing, ARP, ICMP, TCP connections, and application-layer protocols such as HTTP and DNS.

Figure: Wireshark capture showing HTTP packets and detailed Ethernet, IP, TCP and HTTP header information.

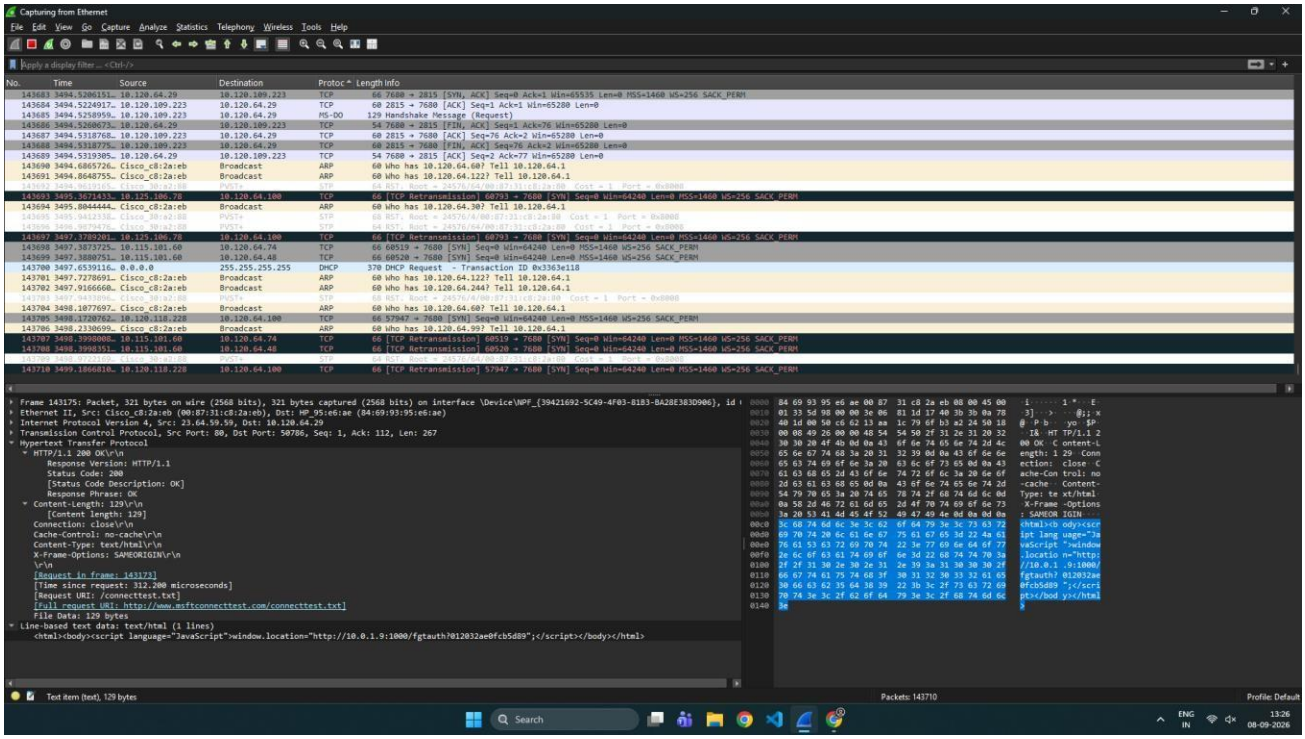


Figure: Wireshark analysis of DNS queries/responses and an ICMP Destination Unreachable message.

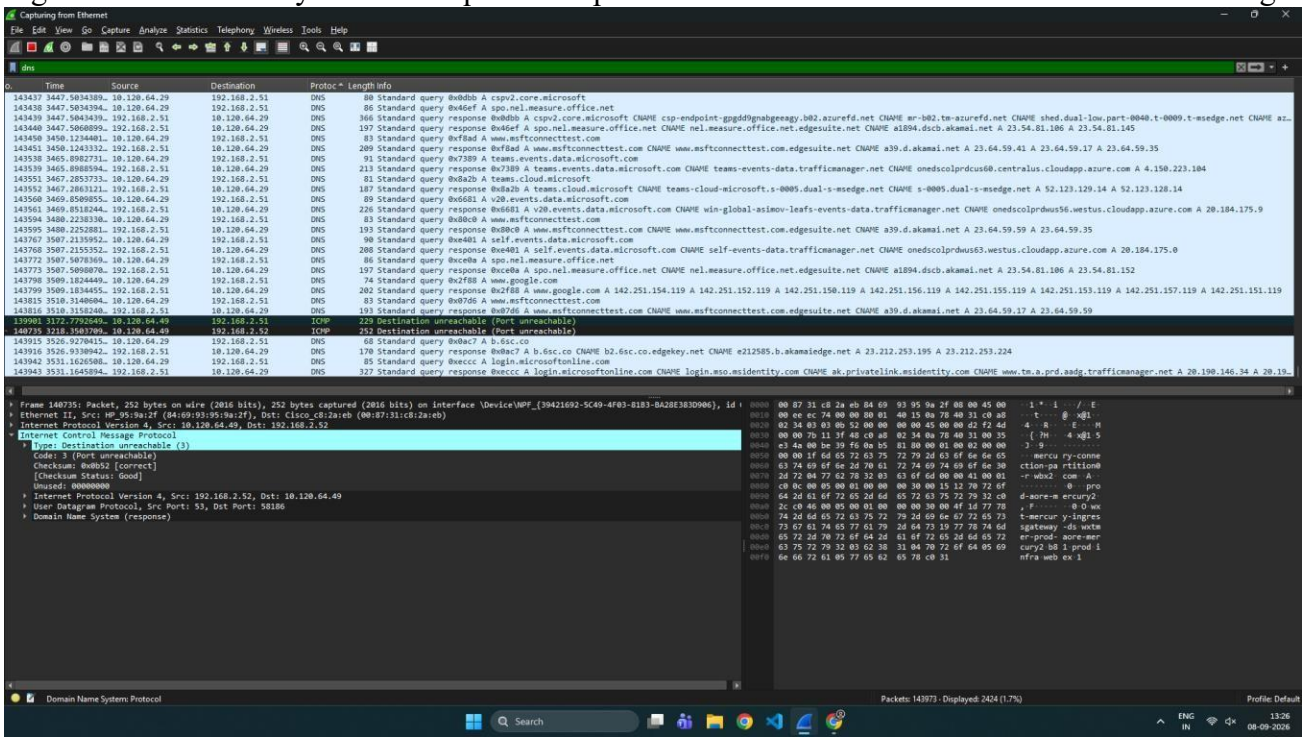


Figure: Following a TCP stream to observe an HTTP GET request and HTTP 200 OK response.

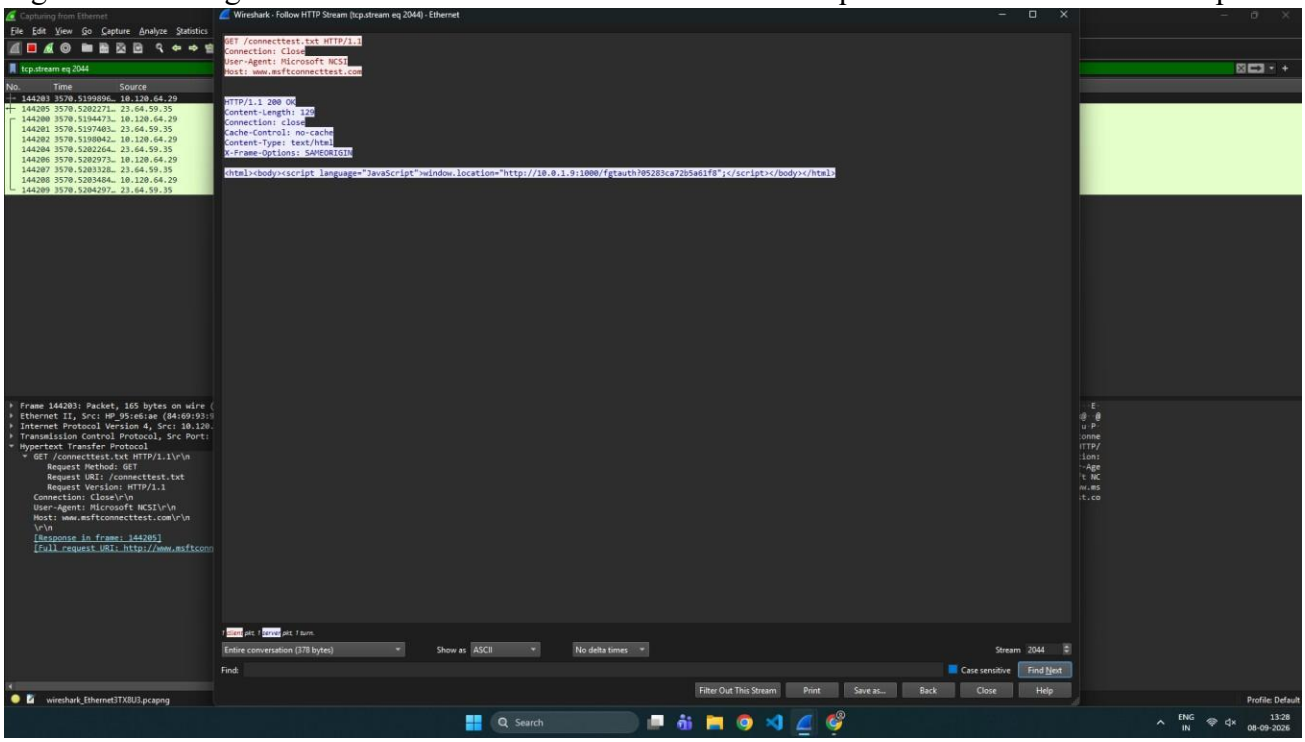


Figure: HTTP GET request and associated HTTP headers captured using Follow TCP Stream.

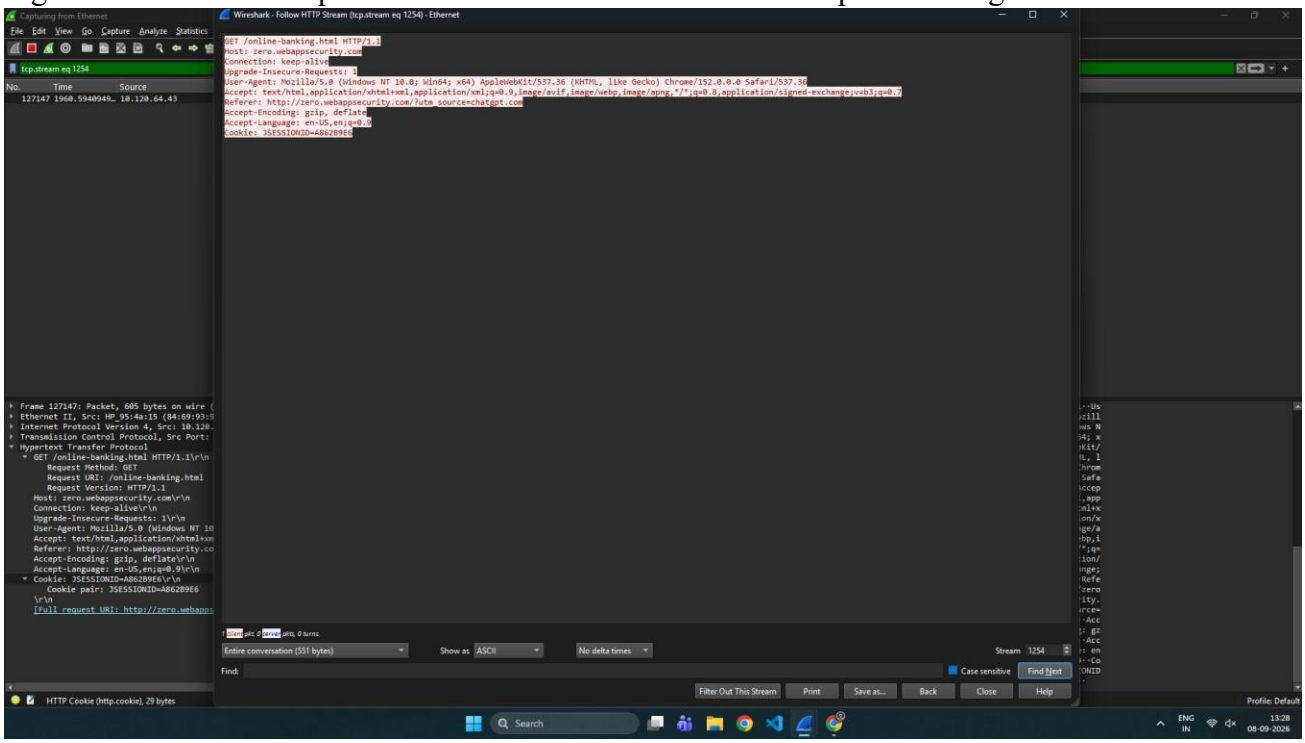


Figure: Detailed analysis of an HTTP GET packet and its protocol headers in Wireshark.

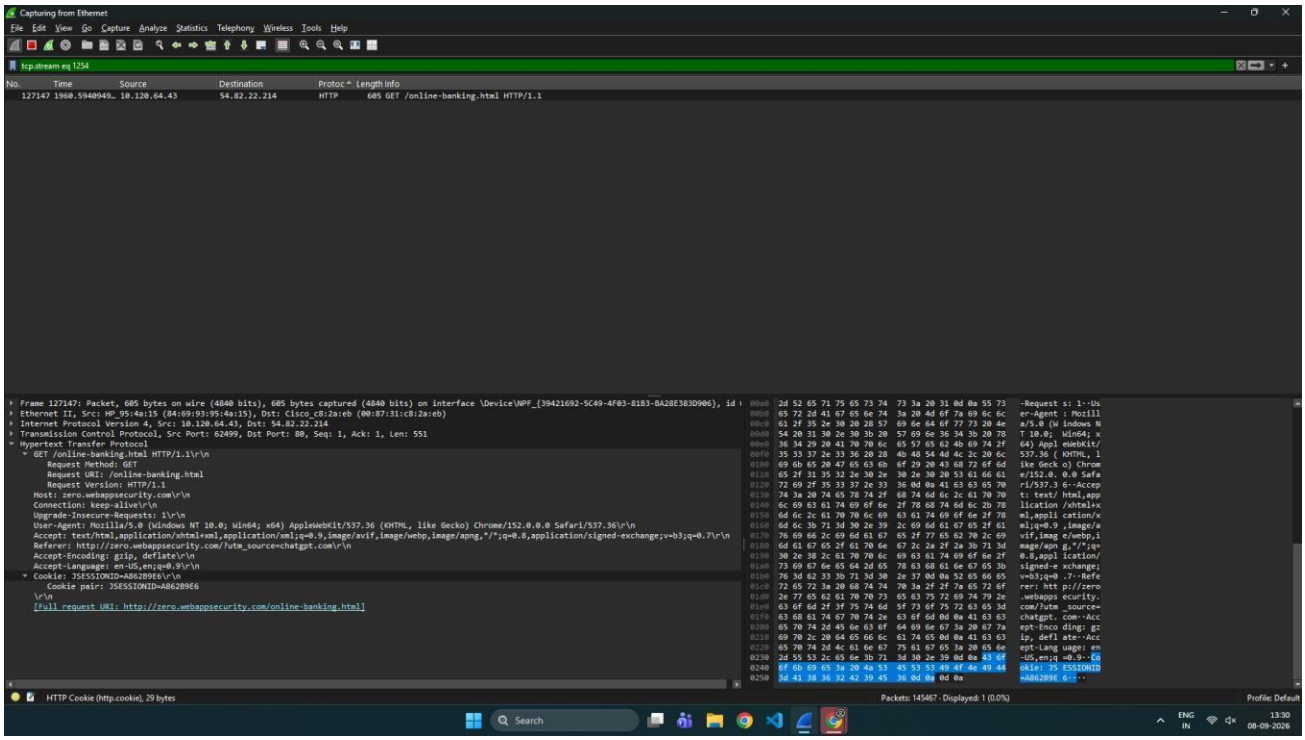


Figure: Wireshark packet capture showing HTTP request-response communication and packet details.

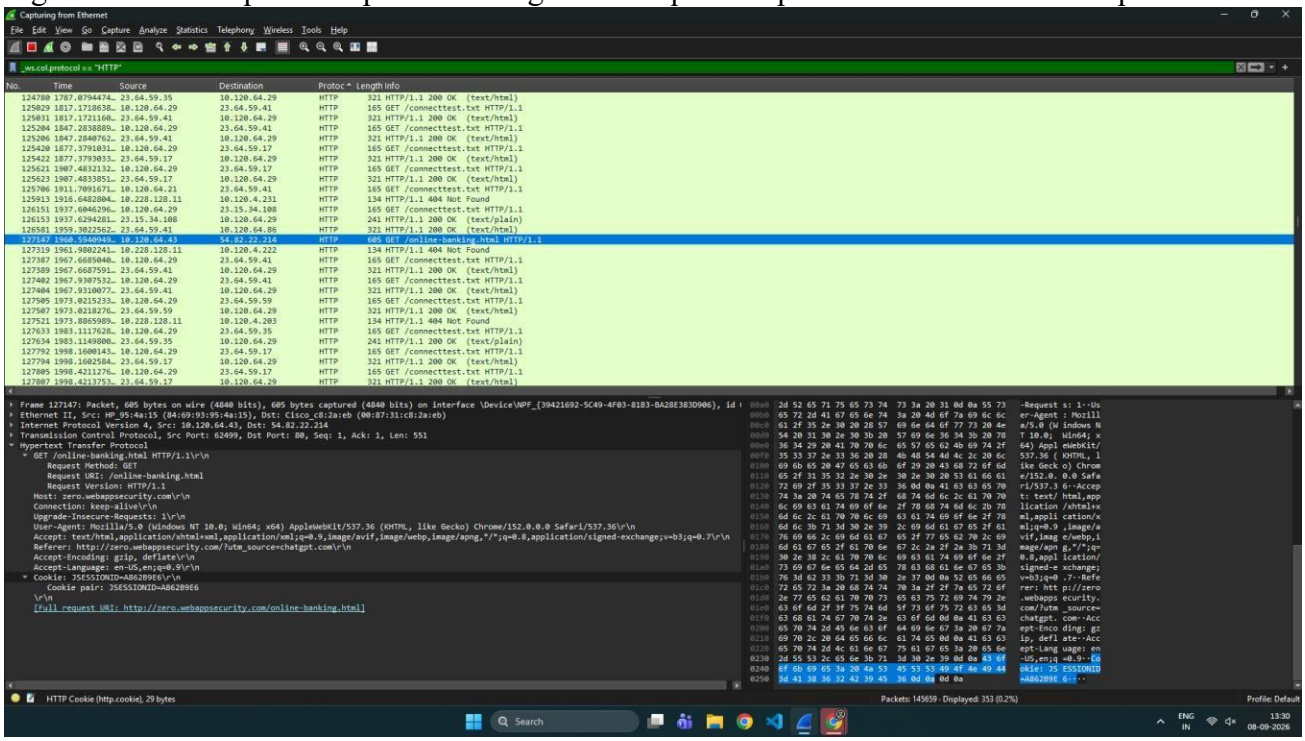
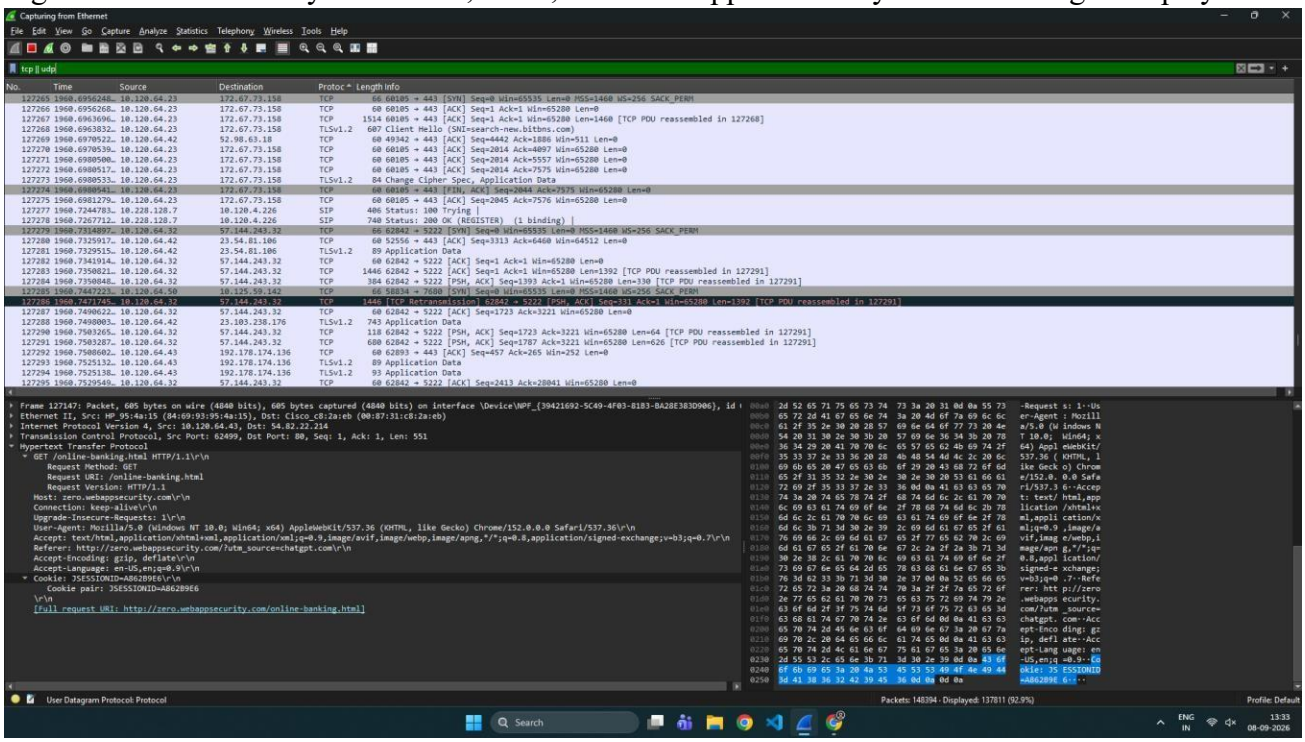


Figure: Wireshark analysis of TCP, UDP, TLS and application-layer traffic using a display filter.



Conclusion:

The experiment successfully demonstrated the use of Wireshark for capturing and analyzing network packets. Various TCP/IP layers and protocols such as Ethernet, ARP, IP, ICMP, TCP, UDP, DNS, HTTP and TLS were observed. The packet details helped in understanding headers, source and destination addresses, ports, requests, responses and protocol encapsulation. The Follow TCP Stream feature further helped in examining the actual communication between a client and server. Thus, Wireshark provides an effective way to understand the practical operation of the TCP/IP protocol stack and network communication.

Reference:

[1] www.geeksforgeeks.org